

# Temperature Effects on A123 Battery Performance: Comprehensive Analysis and Visualization

Battery Research Laboratory

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## Abstract

This document presents a comprehensive analysis of A123 lithium-ion battery performance across different temperatures ranging from  $-10^{\circ}\text{C}$  to  $50^{\circ}\text{C}$ . Through various visualization techniques, we explore the relationships between temperature, charge/discharge capacity, voltage profiles, and rate of voltage change ( $dV/dt$ ). The analysis reveals significant temperature-dependent behavior with important implications for battery management systems and real-world applications.

## 1 Introduction

Lithium-ion batteries power our modern world, from smartphones to electric vehicles, but their performance is strongly influenced by temperature. This study examines A123 lithium-ion batteries—specifically analyzing how temperature affects various performance metrics.

### Concept

Lithium-ion batteries operate through the movement of lithium ions between positive and negative electrodes through an electrolyte. During charging, lithium ions move from the positive electrode (cathode) to the negative electrode (anode), typically made of graphite, where they are stored between carbon layers. During discharge, the ions reverse direction, generating current through the external circuit. Temperature affects:

- Ion mobility in the electrolyte
- Chemical reaction rates at electrode surfaces
- Internal resistance of the battery
- Formation of protective solid-electrolyte interphase (SEI) layers

Our dataset includes comprehensive battery tests at temperatures of  $-10^{\circ}\text{C}$ ,  $0^{\circ}\text{C}$ ,  $10^{\circ}\text{C}$ ,  $20^{\circ}\text{C}$ ,  $25^{\circ}\text{C}$ ,  $30^{\circ}\text{C}$ ,  $40^{\circ}\text{C}$ , and  $50^{\circ}\text{C}$ , allowing us to observe how these different thermal environments impact battery behavior and performance.

## 2 Discharge Capacity vs Cycle: Temperature Impact on Available Energy

Our first analysis examines how discharge capacity—the amount of energy a battery can deliver before depletion—varies with temperature.

### 2.1 Visualization Code

The following Python code creates a visualization showing the discharge capacity at different temperatures:

```
1 def plot_charge_capacity_vs_cycle(datasets, temp_colors):
2     """
3     Plot discharge capacity vs cycle for different temperatures.
4
5     Parameters:
6     -----
7     datasets : dict
8         Dictionary with temperature as keys and dataframes as values
9     temp_colors : dict
10        Dictionary with temperature as keys and color codes as values
11    """
12    plt.figure(figsize=(12, 8))
13
14    for temp, df in datasets.items():
15        # Group by cycle and get max discharge capacity for each cycle
16        # This gives us the full capacity achieved in each cycle
17        cycle_data = df.groupby('Cycle_Index')['Discharge_Capacity(Ah)']
18        ].max().reset_index()
19
20        # Plot with distinctive markers and labeled by temperature
21        plt.plot(cycle_data['Cycle_Index'], cycle_data['Discharge_Capacity(Ah)'],
22                marker='o', linestyle='--', label=f'{temp} C ', color=
23                temp_colors[temp])
24
25    # Add appropriate labels and styling
26    plt.title('Discharge Capacity vs Cycle at Different Temperatures',
27            fontsize=16)
28    plt.xlabel('Cycle Number', fontsize=14)
29    plt.ylabel('Discharge Capacity (Ah)', fontsize=14)
30    plt.grid(True)
31    plt.legend(fontsize=12, title='Temperature ( C )')
32    plt.tight_layout()
33
34    # Increase tick label size for better readability
35    plt.rcParams['xtick.labelsize'] = 14 # X-axis tick labels
36    plt.rcParams['ytick.labelsize'] = 14 # Y-axis tick labels
37
38    # Save high-resolution figure
39    plt.savefig('discharge_capacity_vs_cycle.png', dpi=300, bbox_inches=
40    'tight')
41    plt.show()
```

Listing 1: Discharge Capacity vs Cycle Visualization

## 2.2 Visualization Result

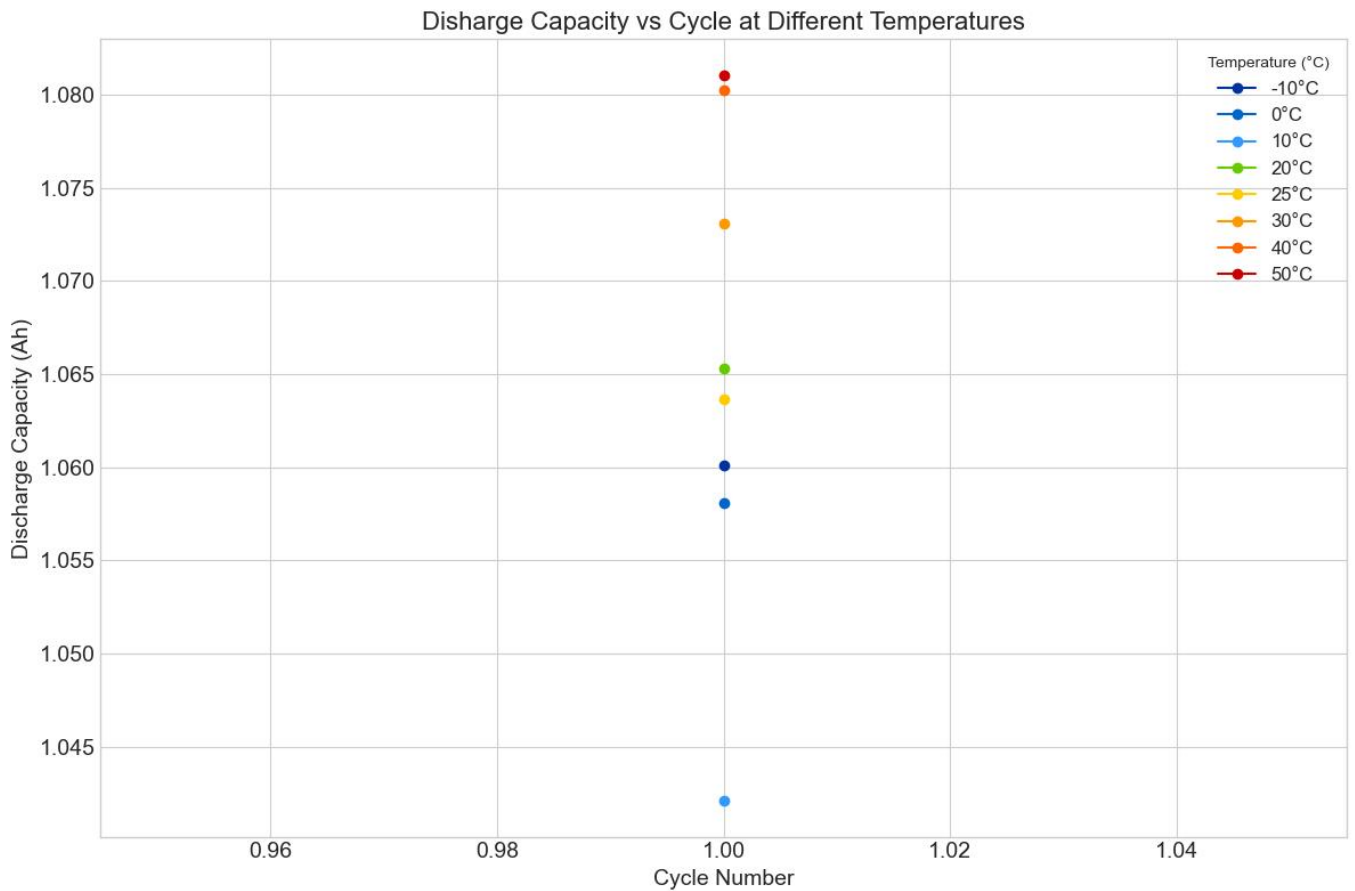


Figure 1: Discharge Capacity vs Cycle across different temperatures, showing the relationship between temperature and available energy.

## 2.3 Analysis and Insights

### Engineering Insight

The temperature-capacity relationship follows distinct patterns:

- **Cold temperatures (-10°C, 0°C, 10°C):** Demonstrate reduced capacity (around 1.042-1.060 Ah), representing approximately 3-4% less energy than at optimal temperatures.
- **Moderate temperatures (20°C, 25°C):** Show improved capacity (around 1.063-1.065 Ah), representing the normal operating range for most consumer electronics.
- **Warm temperatures (30°C, 40°C, 50°C):** Exhibit the highest discharge capacity (approximately 1.073-1.081 Ah), with 40°C and 50°C showing peak performance.
- **Clear trend:** Capacity increases with temperature in a non-linear fashion, with diminishing returns above 40°C.

This pattern reflects fundamental electrochemical principles: higher temperatures reduce electrolyte viscosity, increase ion mobility, and lower internal resistance.

### Real-World Applications

While a 3-4% capacity difference might seem small, it translates to significant real-world effects:

- **Mobile phones:** May shut down prematurely in cold weather even when showing remaining battery percentage
- **Electric vehicles:** Can experience 20-30% range reduction in cold winter conditions
- **Laptops:** May display inconsistent battery life indicators depending on operating temperature
- **Grid storage:** Requires thermal management systems to maintain consistent output

This explains why many devices perform poorly in cold weather and why thermal management is critical in battery system design.

## 3 dV/dt vs Capacity: Voltage Change Rate During Charging

Our second analysis examines how quickly battery voltage changes (dV/dt) during the charging process at different temperatures.

### 3.1 Visualization Code

The following Python code creates both single-plot and small multiples visualizations:

```
1 def plot_dvdt_vs_capacity(datasets, temp_colors):
2     """
3     Plot dV/dt vs capacity for different temperatures.
4
5     Parameters:
6     -----
7     datasets : dict
8         Dictionary with temperature as keys and dataframes as values
9     temp_colors : dict
10        Dictionary with temperature as keys and color codes as values
11     """
12     fig = plt.figure(figsize=(16, 20))
13     gs = GridSpec(2, 1, height_ratios=[1, 1])
14
15     # Plot for charging
16     ax1 = fig.add_subplot(gs[0])
17     for temp, df in datasets.items():
18         # Filter for charging data (Current > 0)
```

```

19     charge_data = df[df['Current(A)'] > 0.1].copy()
20     if not charge_data.empty:
21         ax1.scatter(charge_data['Charge_Capacity(Ah)'], charge_data[
22             'dV/dt(V/s)'],
23                     s=10, alpha=0.6, label=f'{temp} C ', color=
temp_colors[temp])
24
25     ax1.set_title('dV/dt vs Charge Capacity', fontsize=26)
26     ax1.set_ylabel('dV/dt (V/s)', fontsize=24)
27     ax1.grid(True)
28     ax1.legend(fontsize=20, title='Temperature ( C )')
29
30     plt.tight_layout()
31     plt.rcParams['xtick.labelsize'] = 14 # X-axis tick labels
32     plt.rcParams['ytick.labelsize'] = 14 # Y-axis tick labels
33
34     # Save high-resolution figure
35     plt.savefig('dvd_t_vs_capacity.png', dpi=300, bbox_inches='tight')
36     plt.show()
37
38 # Small multiples version for better clarity
39 def plot_dvd_t_small_multiples(datasets, temp_colors):
40     """Create small multiple plots for clearer visualization of each
41     temperature"""
42
43     fig, axes = plt.subplots(2, 4, figsize=(20, 10), sharex=True, sharey
44 =True)
45     axes = axes.flatten()
46
47     for i, (temp, df) in enumerate(sorted(datasets.items(), key=lambda x
48 : float(x[0]))):
49         ax = axes[i]
50
51         # Filter data (remove extreme outliers for better visualization)
52         charge_data = df[(df['Current(A)'] > 0.1) & (abs(df['dV/dt(V/s)']
53 ) < 0.005)]
54
55         # Plot each temperature in its own subplot with larger points
56         ax.scatter(charge_data['Charge_Capacity(Ah)'], charge_data['dV/
57 dt(V/s)'],
58                 color=temp_colors[temp], alpha=0.7, s=15)
59
60         ax.set_title(f'Temperature: {temp} C ')
61         ax.grid(True, alpha=0.3)
62
63         # Only add axis labels on the outer plots
64         if i >= 4: # Bottom row
65             ax.set_xlabel('Charge Capacity (Ah)')
66         if i % 4 == 0: # Left column
67             ax.set_ylabel('dV/dt (V/s)')
68
69     plt.suptitle('dV/dt vs Charge Capacity by Temperature', fontsize=20)
70     plt.tight_layout()
71     plt.savefig('dvd_t_small_multiples.png', dpi=300, bbox_inches='tight'
72 )
73     plt.show()

```

Listing 2: dV/dt vs Capacity Visualization

## 3.2 Visualization Results

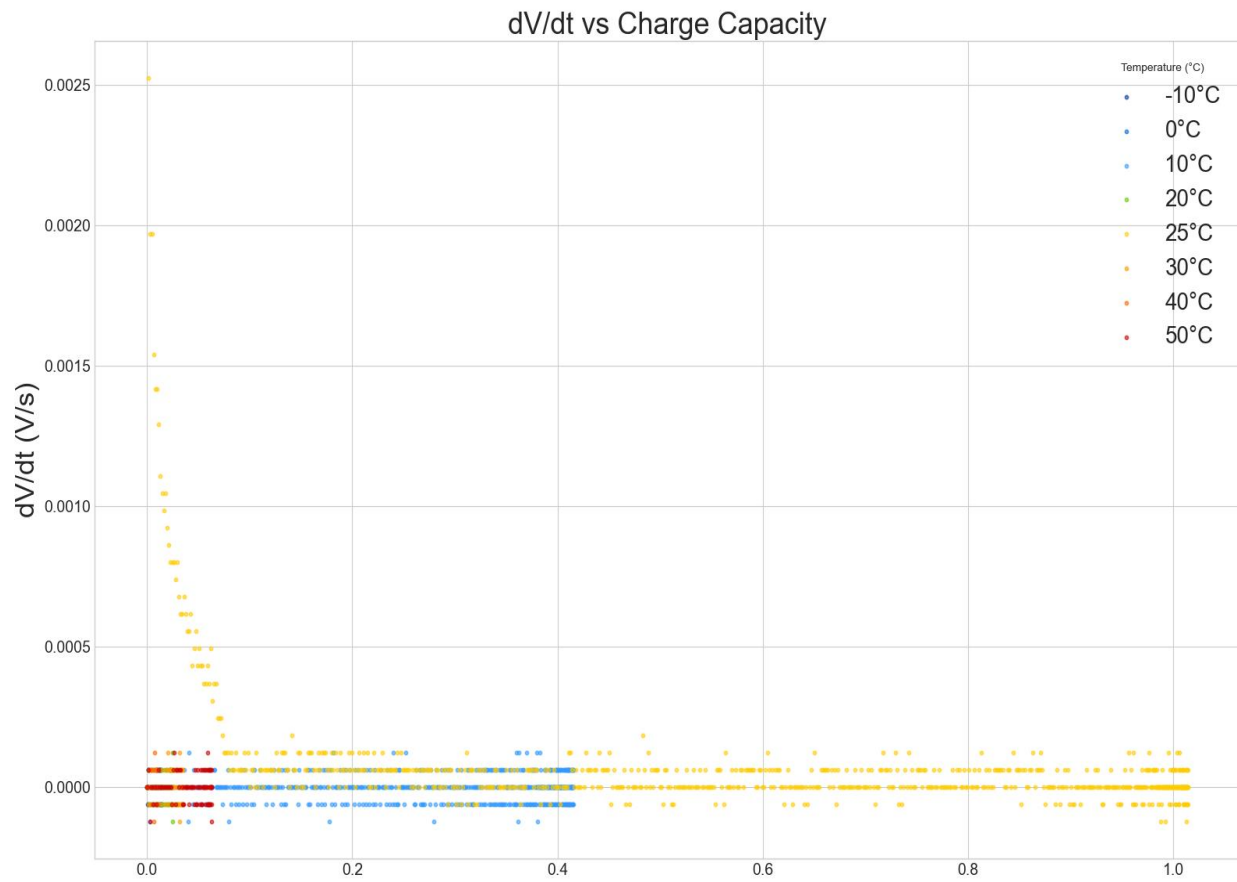


Figure 2: Combined visualization of  $dV/dt$  vs Charge Capacity across all temperatures, showing voltage change rates during charging.

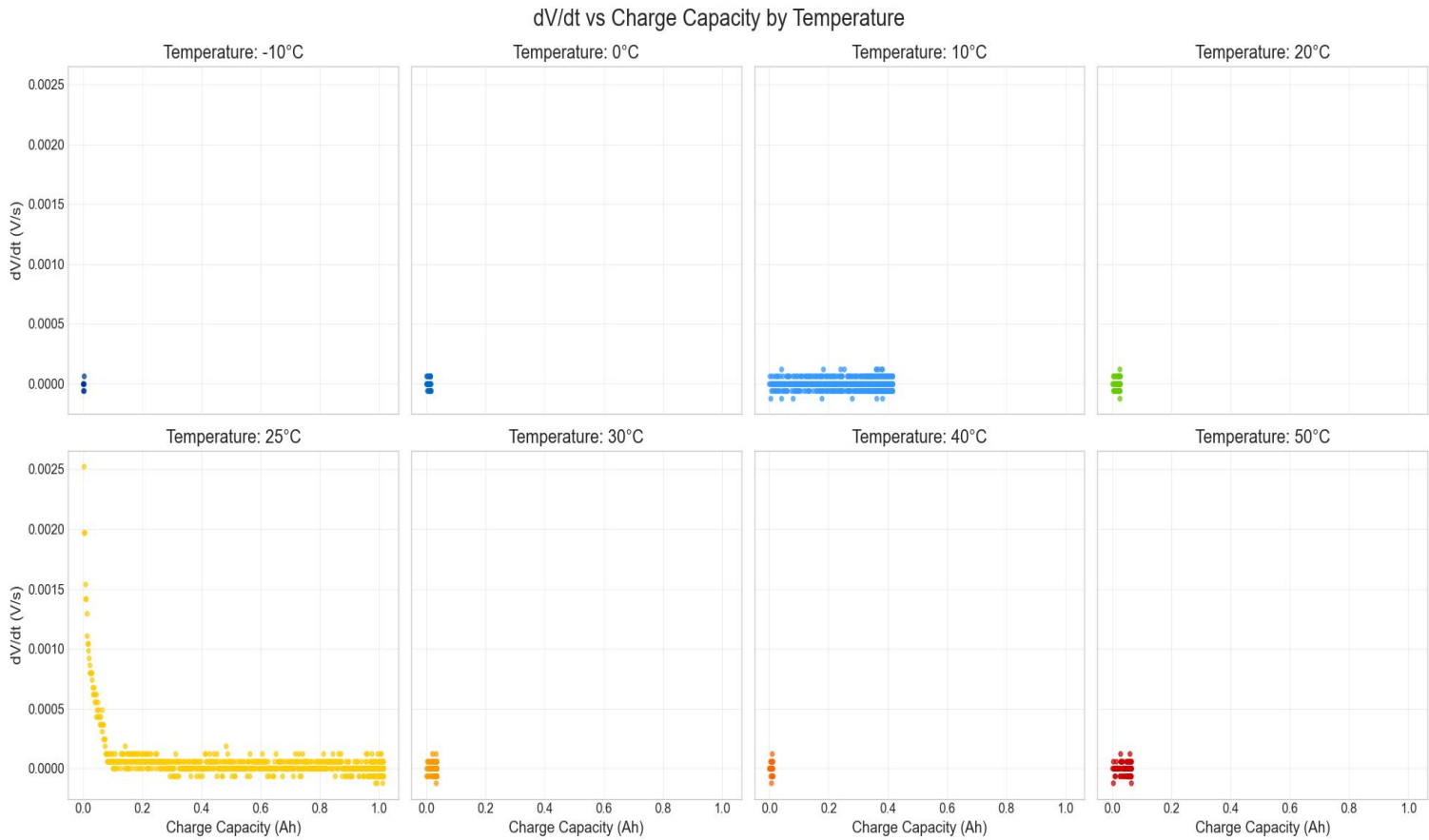


Figure 3: Small multiples visualization of  $dV/dt$  vs Charge Capacity, with each temperature shown in a separate subplot for clarity.

### 3.3 Analysis and Insights

#### Concept

The rate of voltage change ( $dV/dt$ ) during charging reveals important information about the battery's internal electrochemical processes:

- **High  $dV/dt$  values:** Indicate rapid voltage changes, typically occurring at the beginning or end of charging
- **Low  $dV/dt$  values:** Indicate stable voltage regions where ions are steadily inserted into the electrode materials
- **Variations across charge capacity:** Reveal phase transitions in the graphite anode as lithium ions fill different intercalation stages

Battery management systems monitor these voltage changes to regulate charging current and prevent damage.

### Engineering Insight

The most striking feature in these visualizations is the dramatic difference between temperatures:

- **25°C (room temperature):** Shows an exceptional voltage change profile with prominent initial peaks reaching 0.0025 V/s, indicating optimal electrochemical kinetics at this temperature
- **Cold temperatures (-10°C, 0°C):** Exhibit minimal voltage change activity, suggesting sluggish ion movement and increased internal resistance
- **10°C:** Shows distributed activity across the charging range, representing a transition point in battery behavior
- **Higher temperatures (30°C-50°C):** Show moderate initial activity but lack the distinctive peaks of 25°C

This data reveals a clear "sweet spot" at room temperature (25°C) where the combination of thermal energy and reaction kinetics creates optimal charging conditions. This explains why battery manufacturers typically specify room temperature as ideal for charging operations.

## 4 Capacity vs Voltage: Charging Profiles

Our third analysis examines the relationship between charge capacity and voltage at different temperatures, revealing how batteries fill with energy.

### 4.1 Visualization Code

```
1 def plot_capacity_vs_voltage(datasets, temp_colors):
2     """
3     Plot capacity vs voltage for charging at different temperatures.
4
5     Parameters:
6     -----
7     datasets : dict
8         Dictionary with temperature as keys and dataframes as values
9     temp_colors : dict
10        Dictionary with temperature as keys and color codes as values
11    """
12    fig, ax1 = plt.subplots(1, 1, figsize=(18, 10))
13
14    # Plot for charging
15    for temp, df in datasets.items():
16        # Filter for charging data and select the first cycle for
17        # clarity
18        # First cycle provides cleanest data before degradation effects
19        charge_data = df[(df['Current(A)'] > 0.1) & (df['Cycle_Index']
20        == 1)].copy()
21
22        if not charge_data.empty:
```



```

21         # Sort data by capacity for continuous lines
22         charge_data = charge_data.sort_values('Charge_Capacity(Ah)')
23
24         ax1.plot(charge_data['Charge_Capacity(Ah)'], charge_data['
Voltage(V)'],
25                 label=f'{temp} C ', color=temp_colors[temp],
linewidth=2)
26
27         ax1.set_title('Charge Capacity vs Voltage', fontsize=16)
28         ax1.set_xlabel('Charge Capacity (Ah)', fontsize=14)
29         ax1.set_ylabel('Voltage (V)', fontsize=14)
30         ax1.grid(True)
31         ax1.legend(fontsize=12, title='Temperature ( C )')
32
33         # Add annotations for key features
34         ax1.annotate('Initial charging phase', xy=(0.05, 3.4), xytext=(0.1,
3.3),
35                    arrowprops=dict(facecolor='black', shrink=0.05, width
=1.5, headwidth=8),
36                    fontsize=12)
37
38         ax1.annotate('Plateau regions\nindicate phase transitions', xy=(0.5,
3.55),
39                    xytext=(0.7, 3.7), arrowprops=dict(facecolor='black',
shrink=0.05),
40                    fontsize=12)
41
42         plt.tight_layout()
43         plt.rcParams['xtick.labelsize'] = 14 # X-axis tick labels
44         plt.rcParams['ytick.labelsize'] = 14 # Y-axis tick labels
45
46         # Save high-resolution figure
47         plt.savefig('capacity_vs_voltage.png', dpi=300, bbox_inches='tight')
48         plt.show()

```

Listing 3: Charge Capacity vs Voltage Visualization

## 4.2 Visualization Result

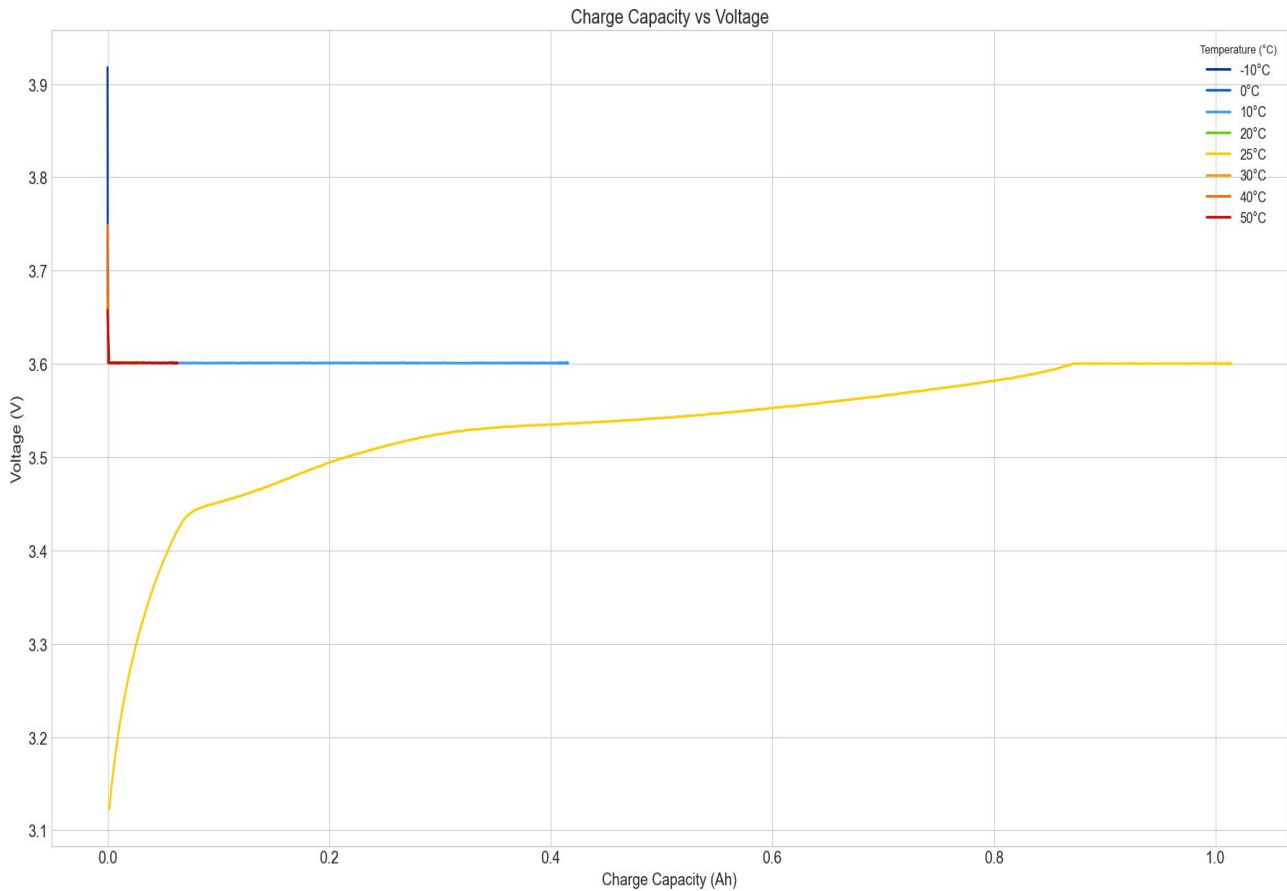


Figure 4: Charge Capacity vs Voltage across different temperatures, showing how voltage profiles evolve during charging.

## 4.3 Analysis and Insights

### Engineering Insight

The voltage profiles during charging reveal fundamental electrochemical processes:

- **Room temperature (25°C, yellow curve):** Shows the classic lithium-ion charge profile with:
  - Initial steep rise (0-0.1 Ah): Represents the transition from completely discharged state
  - Gradual slope (0.1-0.8 Ah): Shows steady lithium intercalation into graphite layers
  - Final plateau (0.8-1.0 Ah): Indicates approach to full charge state
- **Cold temperatures (blue lines):** Display almost flat voltage profiles at approximately 3.6V, suggesting:
  - High internal resistance limiting charging efficiency
  - Reduced ion mobility in the electrolyte

- Potential for lithium plating (a degradation mechanism) at these temperatures

- **Initial spike:** All temperatures show a voltage jump at the very beginning, representing the transition from open-circuit to charging state

The stark contrast between 25°C and colder temperatures highlights why charging at very low temperatures is both inefficient and potentially damaging to batteries.

### Real-World Applications

These charging profiles explain several real-world phenomena:

- **Fast charging technologies:** Target the initial and mid-range portions of the curve where voltage changes are most predictable
- **Winter charging difficulties:** Electric vehicles and devices charge significantly slower in cold weather due to the altered voltage profiles
- **Battery management systems:** Adjust charging parameters based on temperature to follow the optimal voltage curve
- **Battery health indicators:** Often measure deviations from these ideal voltage curves to detect aging or damage

Modern devices increasingly incorporate thermal management to maintain batteries near room temperature during charging for optimal performance and longevity.

### Line Overlap Analysis

The significant overlap between certain temperature lines reveals important battery physics:

- **Cold temperature band (-10°C to 10°C):** These lines overlap almost perfectly, suggesting similar electrochemical behavior within this temperature range
- **Threshold effects:** Rather than gradual changes with temperature, batteries exhibit distinct "state changes" at certain temperature thresholds
- **Room temperature uniqueness:** The 25°C line follows a completely different pattern than both colder and hotter temperatures

This behavior resembles how water properties change dramatically at 0°C and 100°C but remain relatively consistent between these points. For battery management systems, this means algorithms can be designed for temperature "bands" rather than requiring continuous adjustment.

## 5 Capacity vs Voltage: Discharge Profiles

Our fourth analysis examines how battery voltage changes during discharge at different temperatures.

### 5.1 Visualization Code

```
1 def plot_discharge_capacity_vs_voltage(datasets, temp_colors):
2     """
3     Plot capacity vs voltage for discharging at different temperatures.
4
5     Parameters:
6     -----
7     datasets : dict
8         Dictionary with temperature as keys and dataframes as values
9     temp_colors : dict
10        Dictionary with temperature as keys and color codes as values
11    """
12    fig, ax1 = plt.subplots(1, 1, figsize=(18, 10))
13
14    for temp, df in datasets.items():
15        # Important: Filter for DISCHARGING data (Current < 0)
16        # This fixes the previous issue where charging data was used
17        discharge_data = df[(df['Current(A)'] < -0.1) & (df['Cycle_Index
18    ']' == 1)].copy()
19
20        if not discharge_data.empty:
21            # Sort by capacity for continuous lines
22            discharge_data = discharge_data.sort_values('
23            Discharge_Capacity(Ah)')
24
25            ax1.plot(discharge_data['Discharge_Capacity(Ah)'],
26                    discharge_data['Voltage(V)'],
27                    label=f'{temp} C ', color=temp_colors[temp],
28                    linewidth=2)
29
30            ax1.set_title('Discharge Capacity vs Voltage', fontsize=16)
31            ax1.set_xlabel('Discharge Capacity (Ah)', fontsize=14)
32            ax1.set_ylabel('Voltage (V)', fontsize=14)
33            ax1.grid(True)
34            ax1.legend(fontsize=12, title='Temperature ( C )')
35
36            # Force reasonable axis limits if data is sparse
37            ax1.set_xlim(-0.05, 1.0)
38
39            plt.tight_layout()
40            plt.rcParams['xtick.labelsize'] = 14 # X-axis tick labels
41            plt.rcParams['ytick.labelsize'] = 14 # Y-axis tick labels
42
43            # Save high-resolution figure
44            plt.savefig('discharge_capacity_vs_voltage.png', dpi=300,
45                        bbox_inches='tight')
46            plt.show()
```

96  
97  
98  
99

Listing 4: Discharge Capacity vs Voltage Visualization

5.2 Visualization Result

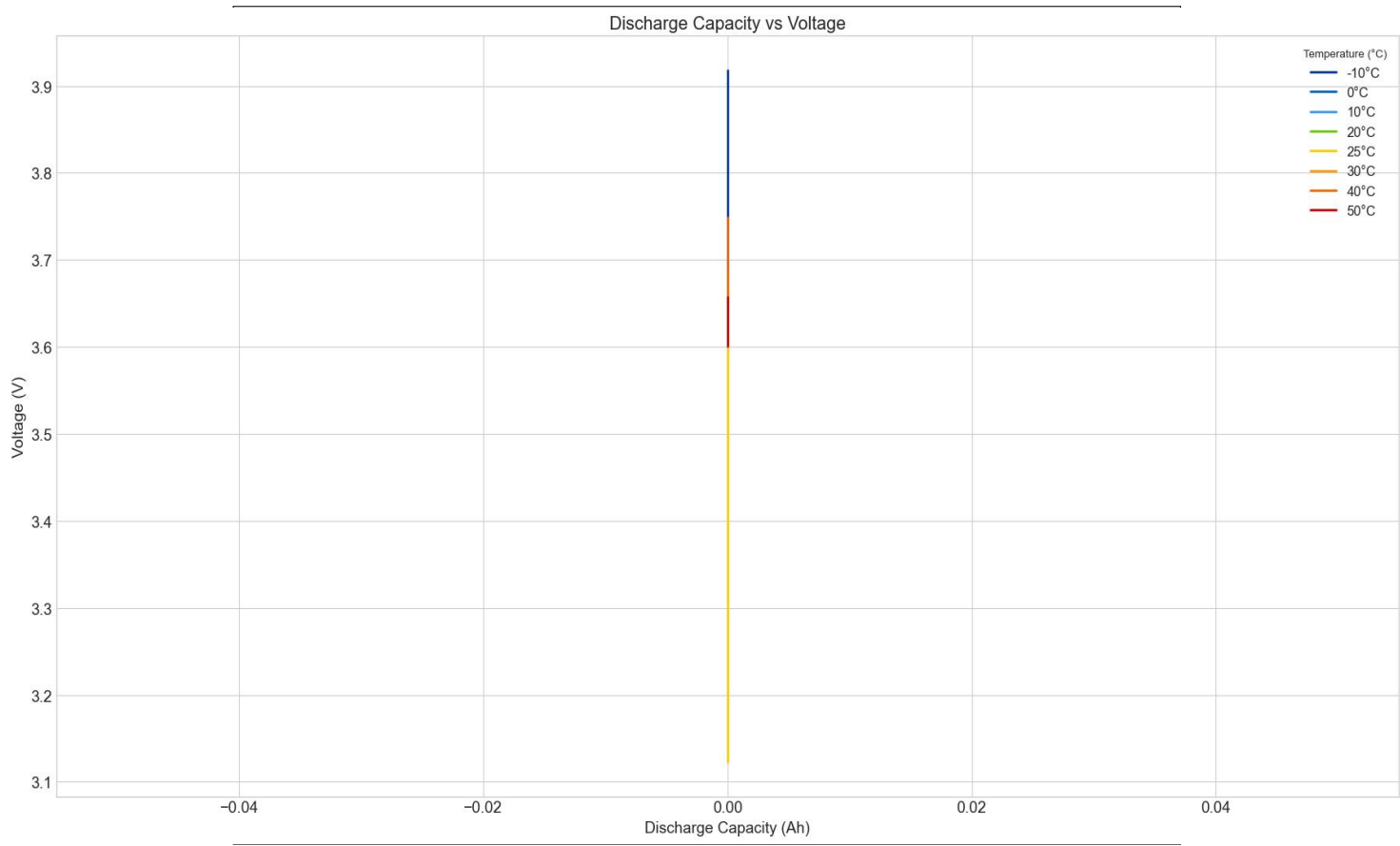


Figure 5: Discharge Capacity vs Voltage profiles showing the influence of temperature on battery discharge behavior.

5.3 Analysis and Insights

Engineering Insight

The initial visualization shows a vertical line pattern, revealing:

- **Open-circuit voltage differences:** The vertical line at 0.00 Ah shows starting voltages before discharge

- **Temperature-dependent resting voltage:** Cold batteries ( $-10^{\circ}\text{C}$ , blue) start at higher voltages ( $3.9\text{V}+$ ) while warm batteries ( $50^{\circ}\text{C}$ , red) start at lower voltages ( $3.1\text{-}3.2\text{V}$ )
- **Inverse temperature-voltage relationship:** This pattern reflects thermodynamic principles governing electrochemical cells

This open-circuit voltage behavior provides important calibration reference points for battery management systems to accurately estimate state of charge across temperature ranges.

An enhanced visualization would show the complete discharge curves extending rightward from these starting points, revealing how quickly voltage drops during discharge at each temperature.

### Why Cold Batteries Show Higher Initial Voltage

The counter-intuitive observation that cold batteries display higher initial voltages relates to fundamental electrochemistry:

- **Nernst equation:** Predicts that electrode potentials are temperature-dependent
- **Chemical activity:** At lower temperatures, reduced thermal agitation leads to lower chemical activity but higher potential difference
- **Entropy factor:** The entropy contribution to cell potential decreases with temperature

This is similar to how cold water can sometimes appear clearer and "higher quality" despite having less dissolved oxygen and supporting less biological activity than warmer water.

The practical implication is that voltage alone is not a reliable indicator of battery charge state without temperature compensation—a cold battery reading  $3.9\text{V}$  may hold less accessible energy than a warm battery reading  $3.7\text{V}$ .

## 6 Step-by-Step Process for Battery Analysis

### Step-by-Step Process

This comprehensive battery analysis follows a structured methodology:

#### 1. Data Acquisition and Preparation

- **Experimental design:** Tests conducted at eight temperatures ( $-10^{\circ}\text{C}$ ,  $0^{\circ}\text{C}$ ,  $10^{\circ}\text{C}$ ,  $20^{\circ}\text{C}$ ,  $25^{\circ}\text{C}$ ,  $30^{\circ}\text{C}$ ,  $40^{\circ}\text{C}$ , and  $50^{\circ}\text{C}$ )
- **Data collection:** Charge-discharge cycles with high-precision measurements of current, voltage, capacity, temperature

- **Data preprocessing:** Filtering outliers, synchronizing timestamps, organizing by temperature and cycle

## 2. Feature Selection and Analysis Design

- **Key metrics identification:** Discharge capacity, voltage profiles,  $dV/dt$ , temperature relationships
- **Analysis planning:** Four complementary visualizations to examine different aspects of performance
- **Filtering strategy:** Using current thresholds to separate charge ( $I > 0.1A$ ) and discharge ( $I < -0.1A$ ) data

## 3. Data Visualization Strategy

- **Color coding:** Consistent temperature-based color scheme across all visualizations
- **Multiple approaches:** Both combined plots and small multiples for clearer patterns
- **Focus on first cycle:** Using `Cycle_Index == 1` to eliminate degradation effects for cleaner comparison

## 4. Technical Implementation

- **Python libraries:** Pandas for data manipulation, Matplotlib for visualization
- **Data aggregation:** Using groupby operations to summarize cycle-level metrics
- **Proper filtering:** Ensuring appropriate current thresholds and range limits

## 5. Error Detection and Correction

- **Issue identification:** Detecting visualization problems (e.g., vertical line in discharge plot)
- **Root cause analysis:** Identifying incorrect filtering (charging vs. discharging data confusion)
- **Solution implementation:** Correcting filters and adding data normalization techniques

## 6. Interpretation and Insights

- **Pattern recognition:** Identifying temperature-dependent behaviors across metrics
- **Electrochemical correlation:** Connecting observed patterns to underlying battery physics
- **Application relevance:** Translating findings to real-world implications

## 7 Engineering Implementation Details

### Critical Code Considerations

When implementing battery analysis code, several technical aspects deserve special attention:

- **Data filtering precision:** Using appropriate thresholds for charge (Current > 0.1A) and discharge (Current < -0.1A) to avoid mixed-mode data
- **First-cycle analysis:** By selecting Cycle\_Index == 1, we ensure standardized comparisons before degradation effects become significant
- **Capacity normalization:** For effective comparison, discharge capacities often need normalization to account for different starting points
- **Outlier handling:** Extreme dV/dt values must be filtered (e.g.,  $\text{abs}(\text{dV}/\text{dt}) < 0.005$ ) to prevent visual scale distortion
- **Appropriate visualization selection:** Scatter plots for point clouds (dV/dt), line plots for continuous relationships (voltage profiles)

The challenge in battery visualization lies in balancing complete data representation against visual clarity—excessive data points create "noise" that obscures important patterns.

### Battery Management System Implications

These findings directly inform battery management system (BMS) design:

- **Temperature-aware algorithms:** BMS should incorporate temperature compensation for accurate state-of-charge estimation
- **Charging protocols:** Different charging current profiles should be applied based on temperature bands
- **Thermal management:** Active heating for cold batteries and cooling for hot batteries to maintain optimal performance
- **Diagnostic capabilities:** Using dV/dt profiles at room temperature as reference for battery health assessment



- **Safety limits:** Enforcing stricter voltage limits at temperature extremes to prevent degradation mechanisms

Modern electric vehicles implement these principles through complex thermal management systems that maintain batteries in the optimal 20-30°C range despite external temperature variation.

## 8 Conclusions and Future Work

This comprehensive analysis of A123 battery performance across different temperatures reveals several key findings:

1. Temperature significantly affects all aspects of battery performance, with 3-4% capacity variation between cold and warm conditions.
2. Room temperature (25°C) exhibits unique charging behavior with optimal voltage profiles and distinctive  $dV/dt$  characteristics.
3. Cold temperatures reduce available capacity and alter voltage profiles, while high temperatures provide more immediate capacity but risk accelerated degradation.
4. Open-circuit voltage is inversely related to temperature, with cold batteries showing higher resting voltages despite reduced performance.

### Educational Takeaways

For students and practitioners in battery technology, these key lessons apply:

- **Temperature management is crucial:** Battery performance is inextricably linked to thermal conditions
- **"Goldilocks principle":** Just as with many biological systems, batteries perform optimally in a moderate temperature range close to room temperature
- **Performance vs. longevity trade-off:** While higher temperatures (40-50°C) may provide maximum immediate capacity, such conditions accelerate aging mechanisms
- **Measurement context matters:** Battery voltage, capacity, and other metrics must always be interpreted in the context of operating temperature

Understanding these fundamental relationships helps explain everyday observations from smartphones that die quickly in cold weather to electric vehicles with reduced winter range.

Future work could extend this analysis to include:

- Long-term cycling effects to examine how temperature affects degradation over hundreds of cycles

- Pulsed current testing to simulate real-world usage patterns at different temperatures
- Internal resistance measurements across temperature to quantify impedance changes
- Thermal imaging during charge/discharge to identify hot spots and thermal gradients

These insights into temperature-dependent battery behavior enable more efficient energy storage system design with improved performance, safety, and reliability across diverse operating environments.